Raffay Hassan

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"Motivated Computer Systems Engineering student with expertise in C++, Python, and modern software developer Proficient in Agile methodologies (XP, CI/CD) and object-oriented design. Experienced in embedded systems, low-code platforms, and cloud integration. Strong in leadership, teamwork, and rapid problem-solving. Currently looking for opportunities to contribute to real-world engineering solutions."

EDUCATION

Middlesex University, London, UK bachelor's in computer systems engineering (Expected Graduation: 2026)

• Achieved First-Class grades in Year 2

• Relevant Coursework: Software Engineering, Digital Systems, Machine Learning, Cybersecurity, Signal Processing, Programming Paradigms, Cloud Computing

PROFESSIONAL EXPERIENCE

IoT-Based Indoor Air Quality Monitoring System [2025]

• Siemens Connected Curriculum Showcase project

• Developed an IoT system using ESP8266, DHT22, and SGP30 sensors to monitor indoor air quality.

• Integrated cloud storage and Mendix for real-time dashboards and alerts.

• Demonstrated skills in embedded C++, cloud APIs, low-code, and OAuth2 setup for Siemens Insights Hub.

Arduino Projects Developer [2023] University Project.

- Developed multiple mini projects using **Arduino**, integrating **sensors** and creating real-world applications with **Python** and **C++**.
- Utilized **GitHub** for version control and collaborated effectively with team members.
- Gained hands-on experience working with hardware, software integration, and real-time systems.

Mechanical Design & CAD Modelling [2023] University Project.

- Utilized **SolidWorks** to create 3D models, simulations, and technical drawings for product designs.
- Optimized designs for manufacturability, improving project efficiency.

Signal Processing [2024] University Project.

- Performed **signal processing** tasks using **MATLAB**, including data filtering, transformation, and analysis for project applications.

FPGA Design and Development [2024]

Designed and developed FPGA circuits using **Xilinx Vivado**, focusing on **digital logic design** and circuit optimization.

ADDITIONAL SKILLS

- **Programming Languages:** C++, Python, Unreal Engine 5 (C++ and Blueprint) , VHDL
- **Cybersecurity:** Ethical Hacking, Vulnerability Analysis, System Protection Techniques
- **Tools & Technologies:** Arduino, Cloud Computing, AI.
- **Software Tools:** SolidWorks (Mechanical Design Projects), Xilinx Vivado (FPGA Development), Altium Designer (PCB Design), MATLAB (Simulation & Modelling), Multisim (Circuit Simulation), Code Composer Studio (Embedded Development) Mendix (low-code app development) , Siemens insights hub.
- **Problem Solving:** Debugging, Performance Optimization
- **Collaboration & Leadership:** Group Leadership, Peer Mentorship, Cross-Functional Teamwork

EXTRA COURSES

- **Intermediate Ethical Hacking and Cybersecurity** (Udemy, in progress).
- **Unreal Engine 5 Game Development (C++ & Blueprint)** (Udemy, in progress.

COMPETITIONS AND ACHIEVEMENTS

- Shortlisted for Siemens Connected Curriculum Industrial Showcase 2025

- Reached National Qualifier stage – WorldSkills UK 2025 (Industrial Electronics)

ADDITIONAL INFORMATION

- **Languages**: English (Fluent) , Urdu (Fluent).
- **Interests**: AI, Cybersecurity, Cloud Computing, Software Engineering, Game Development

PERSONAL STATEMENT

I am a dedicated Computer Systems Engineering student with a strong interest in building real-world digital and embedded systems. I enjoy working on hands-on projects that combine hardware, software, and cloud technologies. My key skills include embedded programming (C++, Python), IoT development with REST API integration, and low-code app design using Mendix. I'm also experienced in using tools like MATLAB, SolidWorks, and Xilinx Vivado, with a solid understanding of Agile workflows and electronics testing.